



Pneumonia Chest X-Ray Classification

A Simple Explanation of the Problem, the Data and the AI Models (PRAICP-1012)

1. What Is Pneumonia, and Why Does It Matter?

Pneumonia is an infection that inflames the tiny air sacs (called alveoli) inside one or both lungs. In a healthy lung, these air sacs fill with air when you breathe. When someone has pneumonia, the air sacs instead fill up with fluid or pus, which makes it harder to breathe and get enough oxygen into the blood.

It is one of the world's leading causes of illness and death, especially in young children and the elderly. Fast, accurate diagnosis matters because the sooner pneumonia is caught, the sooner treatment (usually antibiotics for bacterial cases) can begin.

The standard way doctors check for pneumonia is a chest X-ray. A radiologist looks at the image and searches for cloudy, whitish patches in the lungs, which usually mean fluid has built up where there should be air.

Why Manual Reading Has Limits

Reading X-rays takes a trained radiologist, and there are only so many radiologists available, especially in smaller hospitals or rural clinics. Two different doctors can also sometimes disagree on a borderline image, and reading hundreds of X-rays a day is tiring, which can lead to mistakes. This is the real-world problem this project tries to help with.

2. The Idea: Teach a Computer to Read the X-Ray

This project uses a type of artificial intelligence called a Convolutional Neural Network (CNN) — a kind of deep learning model that is especially good at looking at images. Instead of a person manually pointing out "this part looks cloudy," the CNN is shown thousands of X-rays that are already labelled NORMAL or PNEUMONIA, and it learns, by itself, what patterns of light and dark pixels tend to show up in each case.

Once trained, the model can look at a brand-new X-ray it has never seen before and give a prediction: NORMAL or PNEUMONIA, along with a confidence score. The goal is not to replace the doctor, but to act like a fast first-pass assistant — flagging the images that most likely need urgent attention so the radiologist can review those first.

The Dataset Used

The project used 6,363 real chest X-ray images, already sorted by doctors into NORMAL and PNEUMONIA folders, and split into three groups:

- Training set — 5,218 images used to actually teach the model.
- Validation set — 521 images used to check progress while training.
- Test set — 624 images, kept completely separate, used only at the very end to judge how good the model really is.

One challenge was that the training data had far more PNEUMONIA images (3,876) than NORMAL images (1,342) — almost 3 times as many. If left alone, the model could "cheat" by just guessing PNEUMONIA most of the time and still look fairly accurate. To stop this, the project gave extra importance ("weight") to NORMAL images during training, and also created slightly rotated/shifted copies of training images (data augmentation) so the model would see more variety.

3. Three Different Models Were Built and Compared

Rather than building just one model and hoping for the best, the project trained three different approaches and compared them fairly on the same test images.

Model 1: A Custom CNN (Built From Scratch)

This model was built and trained entirely from zero, learning everything only from the chest X-ray dataset itself. It is like teaching someone to recognise pneumonia patterns having never seen any X-ray, or any picture at all, before.

Model 2 & 3: Transfer Learning (VGG16 and MobileNetV2)

These two models used a shortcut called transfer learning. VGG16 and MobileNetV2 are well-known networks that were already trained on over a million ordinary photographs (animals, objects, people, etc.) to recognise general shapes, edges and textures. Instead of throwing that knowledge away, the project reused it as a head-start, and only trained a small new "decision layer" on top to specialise it for X-rays. This is similar to teaching someone who already knows how to read pictures in general to now specifically recognise pneumonia patterns — much faster and more reliable than starting from nothing.

Fine-Tuning the Best One

After comparing VGG16 and MobileNetV2, MobileNetV2 came out ahead. The project then "fine-tuned" it further — unfreezing a few of its deeper layers and retraining them very gently — to squeeze out a bit more accuracy specifically for X-ray images.

4. How Well Did the Models Actually Do?

To judge the models fairly, the project didn't just look at "percentage correct" (accuracy). In medicine, the more important question is: does the model miss real pneumonia cases? This is measured by a score called Recall — out of all the patients who truly have pneumonia, what percentage did the model correctly catch?

Model	Overall Accuracy	Catches Real Pneumonia (Recall)	Overall Discrimination (ROC-AUC)
Custom CNN (from scratch)	75%	99.7%	0.95
VGG16 (transfer learning)	88%	89%	0.94
MobileNetV2 (transfer learning)	88%	94%	0.95
MobileNetV2 (fine-tuned)	87%	96%	0.96

Simplified results on 624 unseen test X-rays.

The custom CNN almost never misses a real pneumonia case (it catches 99.7% of them) — but it does this by being "trigger-happy": it also wrongly flags a lot of healthy patients as having pneumonia. In fact, out of every 3 truly healthy patients, it only correctly recognises about 1 as healthy. The transfer-learning models are more balanced: they still catch the large majority of real pneumonia cases, while making far fewer false alarms on healthy patients.

Overall, the fine-tuned MobileNetV2 model gave the best all-round balance and was recommended as the most practical option for real use.

5. What This Means in Practice

- An AI model like this can act as a quick, tireless "first look" that flags X-rays most likely to show pneumonia, helping doctors prioritise which patients to review first.
- It should always be used to support a doctor's judgement, never to replace it — it is a screening aid, not a final diagnosis.
- There is always a trade-off between catching every real case (high recall) and avoiding false alarms (high precision); the right balance depends on the hospital and how it plans to use the tool.
- Smaller, faster models like MobileNetV2 are attractive for real-world use because they can run on cheaper hardware, even in clinics with limited resources, while still performing very well.

In short: this project shows that even a moderate-sized set of labelled chest X-rays, combined with modern deep learning techniques, is enough to build a model that can meaningfully assist in pneumonia screening — catching the vast majority of true cases while remaining practical enough to eventually be deployed as a decision-support tool alongside real radiologists.